Feature Selection & LDA

An advanced analysis investigating how different filter feature selection criteria affect the performance of a Multi-Class Linear Discriminant Analysis (LDA) classifier.

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Dataset Description

This assignment utilizes the public **Wine Dataset**. It perfectly satisfies the requirements of having at least 6 features, at least 3 classes, and absolutely no missing values.

13
FEATURES

3
CLASSES

0
MISSING VALUES

178
TOTAL SAMPLES

Top-K Feature Selection (K=5)

Features were ranked using multiple filter ranking metrics. The top 5 features extracted by each criterion to be used for the LDA model are displayed below:

Pearson Correlation

- flavanoids
- od280/od315
- total_phenols
- proline
- hue

Spearman Rank

- flavanoids
- od280/od315
- total_phenols
- hue
- proline

Gini Index

- flavanoids
- color_intensity
- proline
- od280/od315
- alcohol

Chi-Square Statistic

- flavanoids
- color_intensity
- od280/od315
- proline
- hue

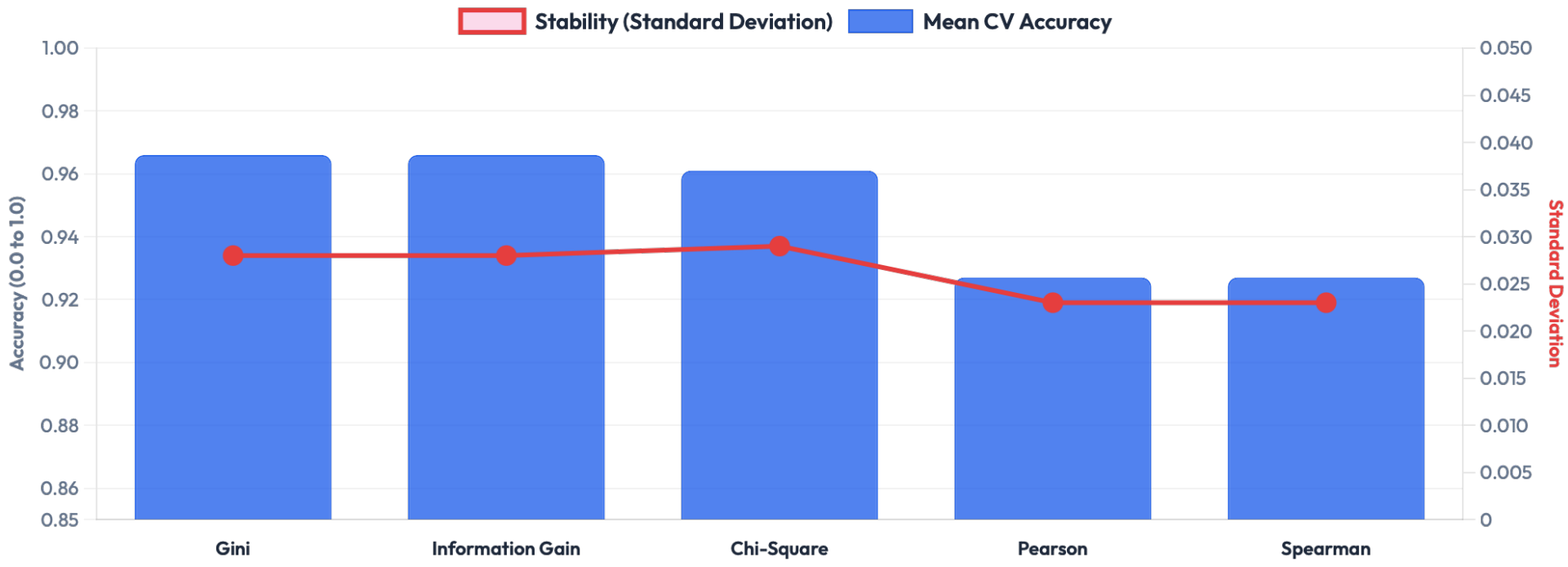
Information Gain

- flavanoids
- od280/od315
- color_intensity
- proline
- alcohol

Performance & Stability Comparison

A 5-Fold Stratified Cross-Validation was run on the Multi-class LDA to evaluate stability (measured by Standard Deviation limits) and performance (Mean Accuracy) across criteria.

METRIC	MEAN CV ACCURACY	CV STANDARD DEVIATION (STABILITY)
Gini Index	**96.60%** (0.966032)	± 0.028062
Information Gain	**96.60%** (0.966032)	± 0.028062
Chi-Square	**96.06%** (0.960635)	± 0.029081
Pearson	**92.68%** (0.926825)	± 0.023114
Spearman	**92.68%** (0.926825)	± 0.023114


Stability (Standard Deviation)
Mean CV Accuracy
Accuracy (0.0 to 1.0)
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0.85
Standard Deviation
0.050
0.045
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0
Gini
Information Gain
Chi-Square
Pearson
Spearman

📖 Theoretical Differences & Analysis

Pearson Correlation

- Assumes a **linear** relationship between continuous features and class labels.
- Highly sensitive to outliers.
- **Pros:** Very fast and simple for predominantly linear dependencies.
- **Cons:** Misses complex, non-linear dependencies.

Spearman Rank Correlation

- Measures **monotonic** relationships through ranks.
- Robust to extreme outliers because it only acts on the relative rank order instead of raw distances.

Gini Index / Impurity

- Evaluates how often a randomly chosen element would be incorrectly labeled.
- Favors continuous features with a large number of distinct outcomes (binning is necessary).
- Excellent for finding local partitions and highly prevalent in Decision Trees.

Chi-Square Statistic

- Compares observed frequencies with theoretically expected ones showing statistical independence.
- Requires categorical / binned data and relies strictly on distribution counts.
- Assumes large enough frequencies per bin for statistical significance.

Information Gain (Entropy-based)

- Represents the reduction in entropy (uncertainty) achieved by knowing a feature's value.
- Inherently unbiased toward pure categorical distributions.
- Tends to capture complex non-linear structures highly effectively compared to linear tests.

Summary Conclusion:
Pearson and Spearman evaluate global linear/monotonic trends. Conversely, Information Gain, Chi-Square, and Gini natively look at structural dependencies via entropy and outcome frequencies, making them suitable when relationships are non-linear or distributions are localized in distinct spaces. This explains the observed performance gap favoring Entropy/Gini methods for this dataset.